外国語科目（数理・計算科学専攻）　　　　　　　　　　　　　１４大修

英語　　　　　　　　　　　　　　　　　　　　　　　時間 午後２時～午後３時

注意事項

1．次の３問中より２問を選んで解答せよ．２問を越えて解答した場合は採点されない可能性がある．

2．解答は１問ごとに別々の解答用紙に記入せよ．

3．各解答用紙には必ず問題の番号および受験番号を記入せよ．

問１．1900 年に Hilbert は数学の未解決問題について歴史的な講演をし，この講演で提起された「Hilbert の問題」は 20 世紀の数学の発展にとても大きく寄与することになった．以下の文章はこの講演の冒頭部分（の英訳）である．

> WHO of us would not be glad to lift the veil behind which the future lies hidden; to cast a glance at the next advances of our science and at the secrets of its development during future centuries? What particular goals will there be toward which the leading mathematical spirits of coming generations will strive? What new methods and new facts in the wide and rich field of mathematical thought will the new centuries disclose?

History teaches the continuity of the development of science. We know that every age has its own problems, which the following age either solves or casts aside as profitless and replaces by new ones. If we would obtain an idea of the probable development of mathematical knowledge in the immediate future, we must let the unsettled questions pass before our minds and look over the problems which the science of to-day sets and whose solution we expect from the future.

（後略）

D. Hilbert "Mathematical Problems" (M.W.Newson 訳)

Bull. Amer. Math. Soc. 8 (1902) 437 より

to-day: today

(1) 枠内の部分を和訳せよ．

(2) 新しい時代をむかえるにあたって，どういう未解決問題を概観することが，どのように意義深いと Hilbert はいっているのか，本文に即して日本語で説明せよ．

問２． 次の英文を読んで，下の（１），（２）に日本語で答えよ．

Once again, opportunity is knocking on the door of the operations research community, this time courtesy of the Internet. Indeed, the upcoming (March-April 2001) issue of Interfaces is devoted to the use of OR with the Internet. (中略) Even IT magazines are beginning to mention OR [Anthes, 2000]. <u>The union of OR and the Internet is a match made in heaven. The reason: the Internet provides inexpensive and ubiquitous communication and data access while OR can support strategic and tactical decision-making with the data.</u>

This is especially true in supply chain management, where collaborative planning and product life-cycle management are just two examples of potential Internet and OR synergy. Companies can use the Internet to share forecasts and production plans with their suppliers and customers, and then use OR-based methods to create optimized plans for the near or medium term. The same synergy can be exploited for product life-cycle management. We can use the Internet to support the collection of customer feedback for use in design and to enable designers to collaborate with marketing, production and suppliers. We can also use OR for improving resource utilization during design and for planning the phase-in and phase-out of products.

ManMohan S. Sodhi, "A Match Made in Heaven",
ORMS Today, Vol. 28, No. 1 (2001)より

Interfaces : 雑誌名
OR: Operations Research
synergy: Cooperative interaction among groups that creates an enhanced combined effect

(1) ＿＿部分を和訳せよ．
(2) The union of OR and the Internet は supply chain management にどのようなメリットをもたらすか，本文の内容に即して説明せよ．

問３．以下の文章の内容に即して以下の設問に日本語で答えよ．

In addition to signed and unsigned integers, programming languages support numbers with fractions, which are called *reals* in mathematics. Here are some examples of reals:

$3.14159265\ldots_{ten}$ (π)

$2.71828\ldots_{ten}$ (e)

0.000000001_{ten} or $1.0_{ten} \times 10^{-9}$ (seconds in a nanosecond)

$3{,}155{,}760{,}000_{ten}$ or $3.15576_{ten} \times 10^{9}$ (seconds in a typical century)

Notice that in the last case, the number didn't represent a small fraction, but it was bigger than we could represent with a 32-bit signed integer. The alternative notation for the last two numbers is called *scientific notation*, which has a single digit to the left of the decimal point. A number in scientific notation that has no leading 0s is called a *normalized* number, which is the usual way to write it. For example, $1.0_{ten} \times 10^{-9}$ is in normalized scientific notation, but $0.1_{ten} \times 10^{-8}$ and $10.0_{ten} \times 10^{-10}$ are not.

Just as we can show decimal numbers in scientific notation, we can also show binary numbers in scientific notation:

$1.0_{two} \times 2^{-1}$

To keep a binary number in normalized form, we need a base that we can increase or decrease by exactly the number of bits the number must be shifted to have one nonzero digit to the left of the decimal point. Only a base of 2 fulfills our need. Since the base is not 10, we also need a new name for decimal point; *binary point* will do fine.

Computer arithmetic that supports such numbers is called *floating point* because it represents numbers in which the binary point is not fixed, as it is for integers. The programming language C uses the name *float* for such numbers. Just as in scientific notation, numbers are represented as a single nonzero digit to the left of the binary point. In binary, the form is

$1.xxxxxxxxx_{two} \times 2^{yyyy}$

(Although the computer represents the exponent in base 2 as well as the rest of the number, to simplify the notation we'll show the exponent in decimal.)

A standard scientific notation for reals in normalized form offers three advantages. It simplifies exchange of data that includes floating-point numbers; it simplifies the floating-point arithmetic algorithms to know that numbers will always be in this form; and it increases the accuracy of the numbers that can be stored in a word, since the unnecessary leading 0s are replaced by real digits to the right of the binary point.

D.A.Patterson & J.L.Hennessy "Computer Organization & Design" (2nd Ed.) より

(1) "normalized scientific notation" とは何か答えよ．また，次の(a), (b) はそれぞれ normalized scientific notation であるか否か，理由とともに答えよ．

(a) $31.4_{ten} \times 10^{-1}$

(b) $0.11_{two} \times 2^{3}$

(2) "scientific notation for reals in normalized form" が，計算機が扱う数値の精度を高めるのはなぜか，理由を答えよ．

注： 3.14_{ten} の ten は，その数が１０進表記であること，0.11_{two} の two は，その数が２進表記であることを表す．